

The drilling speed $s(\chi)$: a lemma candidate

Denis Joubert

Independent researcher, Switzerland

Grok

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September 1, 2026 — lemma campaign, item 2

Abstract

The affine law $-\ln \lambda_{\min} = s(\chi)\mu + b(\chi)$ factorizes the depth problem into a window-invariant staircase $\Delta(\ell)$ and one arithmetic speed $s(\chi)$. The measured four-variable map for s on the Dirichlet family is recorded, the ζ -outlier is isolated as the pole, and two assertions a derivation must supply are stated: (S1) linearity in μ is the prime-number theorem read on the radical; (S2) the prefactor $s(\chi)$ is the Lyapunov exponent of windowed Fourier restriction into the desert, equal to the per-digit cost of reading γ_1 (precision law). Three preregistered predictions for the missing characters $\chi_{-8}, \chi_{-20}, \chi_{-23}$ are attached. Nothing is claimed about RH.

1 What is already measured

On fourteen real primitive characters of conductor ≤ 24 , at three windows each [?],

$$-\ln \lambda_{\min}(Q_{\chi,\mu}) = s(\chi)\mu + b(\chi),$$

affine, with s ranging from 0.46 (χ_{24}^{odd}) to 4.00 (χ_3). ζ at a depth-adequate basis gives $s_\zeta = 11.7$, off the Dirichlet family by a factor three at the boundary of the map. The precision campaign [?] §16.3 then identified s a second time: the MUSIC error on γ_1 falls as $e^{-s\mu}$, not $e^{-s\mu/2}$. So $s(\chi)$ is the per-digit cost of reading the first zero from the primes.

The four-variable map on the Dirichlet family (leave-one-out 4.8%, RMS 6.1%; ζ excluded) is

$$\hat{s} \approx 0.14 \cdot \gamma_1^{1.32} (\gamma_2 - \gamma_1)^{0.45} e^{-0.13D} \cdot 1.32^{[\text{odd}]},$$

with $D = \sum_{p|q} \log p / (\sqrt{p} - 1)$ the arithmetical mass removed. Residuals of χ_{12} (+14%) and χ_{13} (−10%) remain open.

2 The lemma candidate

Conjecture 1 (Drilling speed). *There is a functional s of the pair $(\chi, \text{window construction})$ such that $\ell_0(\chi, \mu) = s(\chi)\mu + b(\chi) + o(1)$ as μ grows through depth-adequate bases. The functional factorizes as*

$$s(\chi) = \kappa_{\text{win}} \cdot \Lambda(\chi),$$

where κ_{win} is a window invariant (the same for every χ) and $\Lambda(\chi)$ is an arithmetic coupling of the desert $(\gamma_1, \gamma_2 - \gamma_1)$, the conductor mass D , and the parity.

Together with the universality conjecture for $\Delta(\ell)$ this reproduces the whole quasi-null ladder from two arithmetic scalars and one window profile.

3 What a derivation must supply

S1. Linearity is the prime-number theorem on the radical. Each new prime power $p^k \leq \mu$ reinforces the rungs that already recruited p , and $\psi(\mu) \sim \mu$ counts those powers. Hence $d\ell_0/d\mu$ is $O(1)$, not $O(1/\mu)$ (continuum / Paley–Wiener) and not $O(\log \mu/\mu)$ (Landau count on $\{\log n\}$). Continuum analysis of the window cannot produce $s\mu$. The lattice is mandatory.

S2. s is the Lyapunov exponent of the desert. Hyper-nullity says $|\hat{v}_0(\gamma_1)| \ll \sqrt{\lambda_0}$: interior zeros are cancelled far past the L^2 budget, and the residual mass sits at the band frontier. Differentiating the implicit relation $Q(v_0) = \lambda_0$ along μ then yields

$$s = -\frac{d}{d\mu} \log |\hat{v}_0(\gamma_1)| + o(1),$$

which is the precision law already measured. Deriving the four-variable map is then deriving how $(\gamma_1, \text{gap}, D, \text{parity})$ set that logarithmic derivative. The map is the phenomenological shadow of S2, not a substitute for it.

The ζ -outlier is S1+S2 with an extra rank-one pole. Removing the pole collapses exactly one direction and leaves the rest of the ladder intact [?] §14.5; s_ζ is therefore not the Dirichlet map evaluated at $\gamma_1(\zeta) = 14.13$.

4 Preregistered predictions

Zeros harvested 1 September 2026 (completed Λ -function, bisected; caches `zeros_chim8.pkl`, `zeros_chi20.pkl`, `zeros_chi23.pkl`). The map gives

χ	q	γ_1	$\gamma_2 - \gamma_1$	D	$\hat{s}$
χ_{-8}	8	3.576	3.858	1.673	1.47
χ_{-20}	20	2.359	2.317	2.975	0.57
χ_{-23}	23	2.871	1.344	0.826	0.76

A measured s off $\hat{s}$ by more than 20% on any of the three kills the four-variable map as stated. χ_{-23} remains the sharpest test: small gap, low D , odd, class number 3.

5 Status

Conjecture ?? is the lemma candidate. S1 and S2 are the proof obligations. The four-variable map is a measurement, not the lemma. The three predictions above are the cheapest falsifiers.

References

- [1] D. Joubert, *Depth phenomenology of truncated Weil quadratic forms*, same repository (2026).
- [2] D. Joubert, *Le milieu des nombres premiers*, lab notebook, same repository (2026).

Addendum — endpoint law, 1 September afternoon

Notebook §16.7 (now on `main`) supplies the missing identification of S2. Across four configurations and sixty orders of magnitude,

$$|\hat{v}_0(\gamma_1)| \approx C \cdot \lambda_0, \quad C \in [7, 25].$$

That is why the MUSIC error tracks $e^{-s\mu}$ and not $e^{-s\mu/2}$, and why the leakage slope on ζ satisfies $\tau \approx s\mu/(2\omega_{\max})$ to 6%. D2 is dead: τ depends on μ . s remains the primary arithmetic speed; τ is its projection onto the frequency axis at a given window.

Addendum — held-out slopes, 1 September

`code/scan_s.py`, last segments: χ_{-8} : $s = 1.46$ against $\hat{s} = 1.47$ (-0.5%); χ_{-20} : $s = 0.55$ against 0.57 (-3.7%); χ_{-23} : $s = 0.47$ against 0.76 (-38%), slope stable from $\mu = 16$ to 32 . Two confirmations and a kill. S2 is identified by the endpoint law; the four-variable map is not S2, and χ_{-23} just said so.